2020-2021 SPRING SEMESTER FLUID MECHANICS FINAL EXAM, Date:15/06/2021

Name: **Surname:** **Student ID:**

Question 4 (25p):

a) The vertical distance traveled (Δz) per unit time (Δt) of a falling object in the air is assumed to depend on the density of the object (ρ) and the gravitational acceleration (g). Obtain an expression for Δz as a function of non-dimensional groups with the aid of dimensional analysis.

b) For an incompressible, two-dimensional, continuous and irrotational flow field, the velocity component in the x-direction is given by the equation u= $\frac{y^3}{3} - yx^2 - (a+b) * x$.

a) Determine the velocity component in the y-direction.

b) Obtain the expression for the velocity potential.

c) Obtain the expression for the stream function.

a, and b are the last two digits of your student ID.
For example, Student ID: 17042082 a=8 b=2